Factors Affecting Conscientiousness

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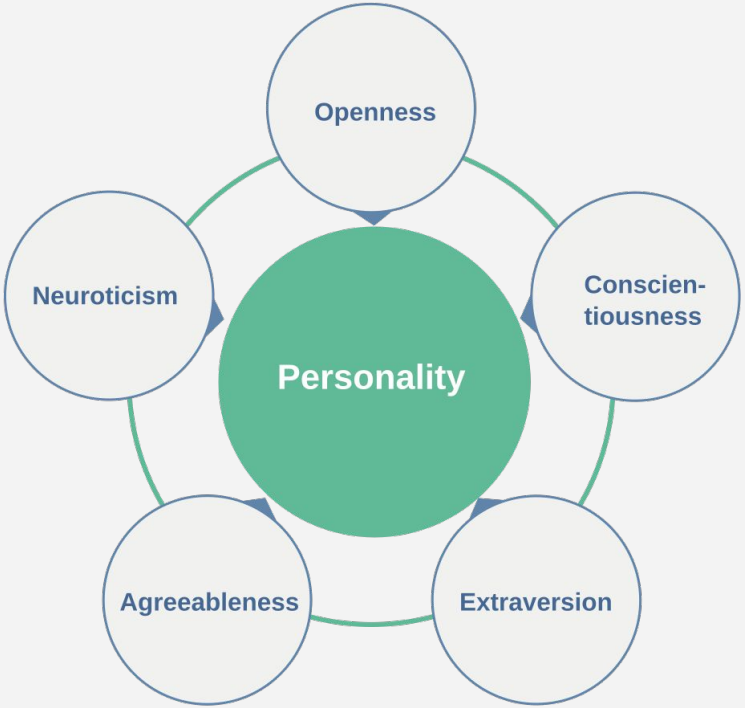
What is conscientiousness?

One of the Big Five personality traits

Set of personality constructs related an individual's propensity to be self-controlled, responsible to others, hardworking, orderly, and rule-abiding.

High levels of conscientiousness are associated with:

- High achievement in formal education
- High job performance
- Marital stability
- Lower risk of depression



(Roberts et al., 2014); image: Anna Tunikova

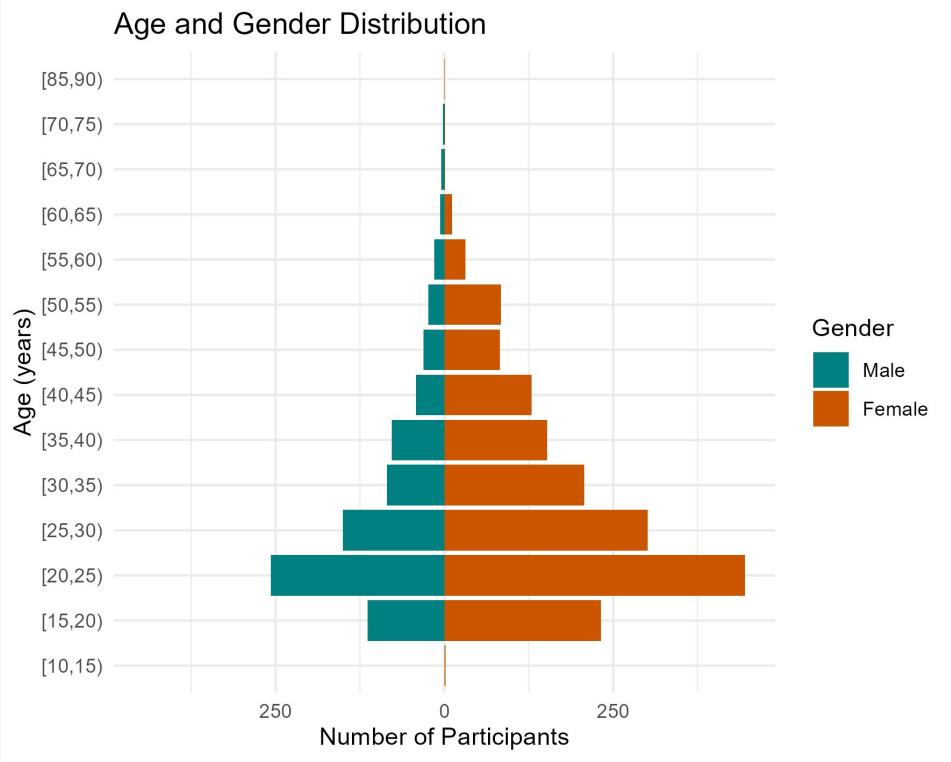
Participants: Age & Gender

Responses:

- Males: 809
- Females: 1679
- Total: 2488

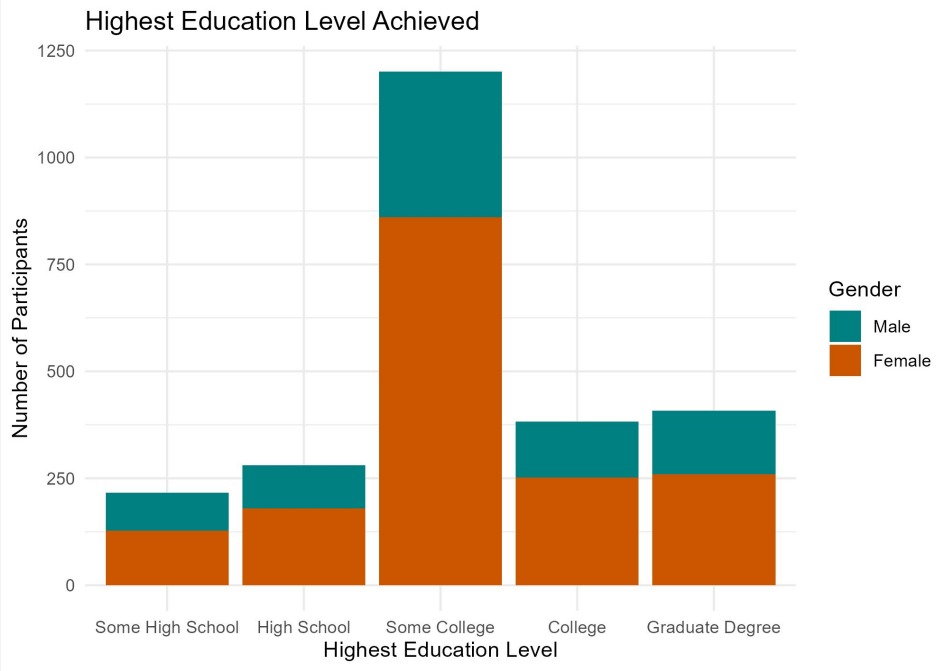
Age (years):

- Combined Median: 26
  - Males Median: 25
  - Females Median: 27
- Youngest: 14
  - Males: 15
  - Females: 14
- Oldest: 86
  - Males: 74
  - Females: 86



Participants: Education

- Some High School:
  - Male: 88
  - Female: 128
- High School:
  - Male: 101
  - Female: 180
- <u>Some College:</u>
  - Male: 341
  - Female: 860
- College:
  - Male: 130
  - Female: 252
- Graduate Degree:
  - Male: 149
  - Female 259



Conscientiousness Questionnaire

Questions 4 and 5 are reverse scored

Maximum score of 24

Score	Response
NA	NA
1	Very Inaccurate
2	Moderately Inaccurate
3	Slightly Inaccurate
4	Slightly Accurate
5	Moderately Accurate
6	Very Accurate

Questions:

1. Am exacting in my work.
2. Continue until everything is perfect.
3. Do things according to a plan.
4. Do things in a half-way manner.
5. Waste my time.

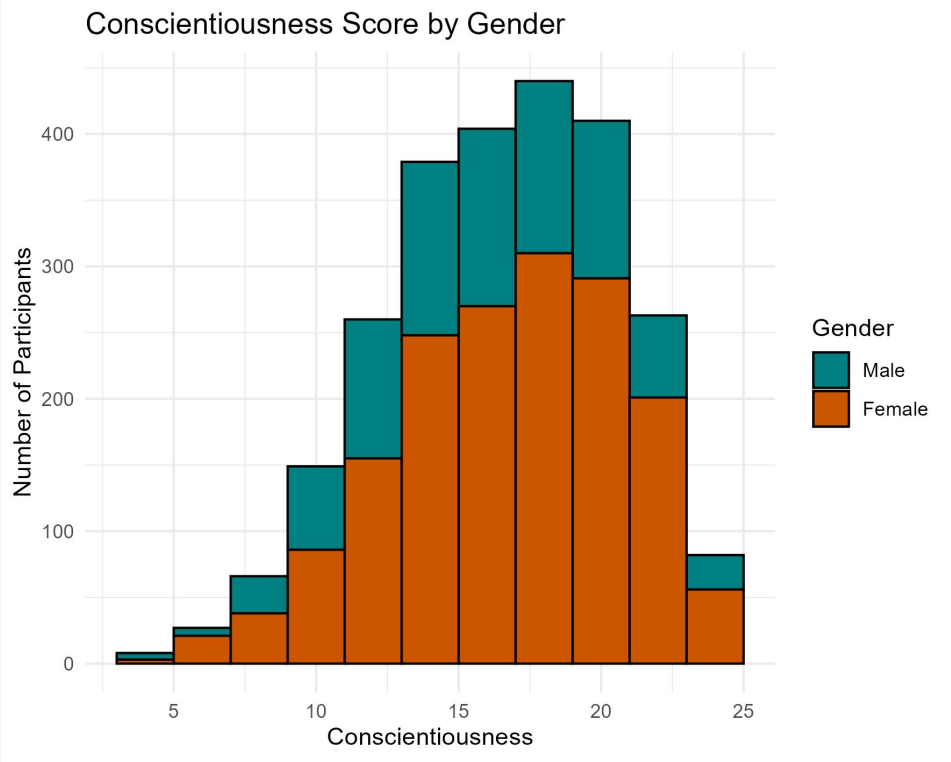
Questionnaire Results

Average Score Overall = 16.98

- Average Male = 16.36
- Average Female = 17.24

Reliability Measures:

- α = 0.72, 95% CI [0.70, 0.74]
- $\square_h$ = 0.73, 95% CI [0.71, 0.74]



Factors Affecting Conscientiousness

Age is positively correlated showing a small but statistically significant effect.

Gender shows a significant effect with females averaging 0.78 higher on our conscientiousness scale.

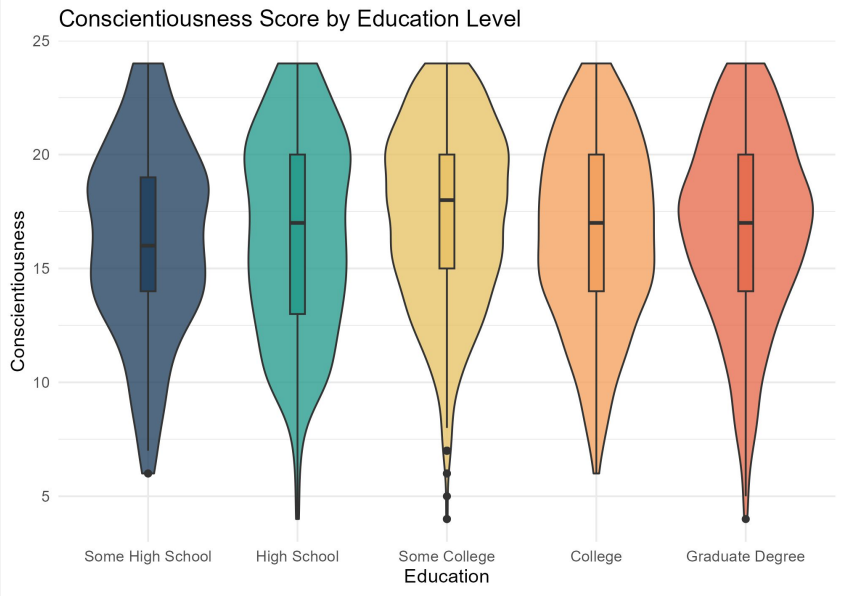
Some college education is associated with an average of 0.98 higher scores compared to some high school education. All other education levels showed no significant difference from some high school education.

Characteristic	Beta	95% CI	p-value
Age	0.04	0.02, 0.05	**<0.001**
Gender			
Male	—	—	
Female	0.76	0.42, 1.1	**<0.001**
Education			
Some High School	—	—	
High School	-0.02	-0.74, 0.69	>0.9
Some College	0.90	0.32, 1.5	**0.002**
College	-0.10	-0.78, 0.58	0.8
Graduate Degree	0.13	-0.55, 0.81	0.7
R^2	0.029		
Adjusted R^2	0.026		
No. Obs.	2,488		
Statistic	12.2		
p-value	<0.001		

Abbreviation: CI = Confidence Interval

Summary of Findings

- Conscientiousness increases with age
- Females are more conscientious than Males
- Participants with some college education showed higher conscientiousness scores than those with higher education levels. This is counter to what would be expected as conscientiousness is considered a predictor of higher-education success and thus you would expect to find higher levels of this personality trait present in the higher education population (Roberts et al., 2014).



Appendix

2800 survey responses total → 2488 remain after cleaning:

- 223 removed for NA in education
- 2 removed for age <14
- 87 removed for NA in survey questions

No interaction between predictor terms:

- VIF <1.2 for all terms

References

Roberts, B. W., Lejuez, C., Krueger, R. F., Richards, J. M., & Hill, P. L. (2014). What is conscientiousness and how can it be assessed?. *Developmental psychology*, *50*(5), 1315.

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